

CAS Assignment - 3

Name - Chinmayee Rana
Regd. No. - 2141018149
Section - CSE - 'Z'

Trends and Cloud Services Overview:—

Serverless Architecture:— Lets developers run code without managing servers. Platforms like AWS Lambda automatically scale and only charge for usage. It suits event-driven apps like APIs or file processing, reducing infrastructure concerns & speeding up development.

Progressive Web Apps (PWAs):— blend web and mobile app features. They work offline, send push notifications, and can be installed on a home screen. With service workers and caching, PWAs provide a smooth, app-like experience.

AI & Machine Learning:— enhance applications with intelligence. Use cases include personalization, fraud detection & analytics. Architectures now include ML pipelines, real-time inference, & scalable data flows, often integrated with cloud tools like AWS SageMaker or Azure AI services.

Cloud Models:—

- SaaS: full software solutions (e.g., Zoom)
- PaaS: Platforms for app development (e.g., Google App Engine)
- IaaS: on-demand infrastructure (e.g., AWS EC2)